

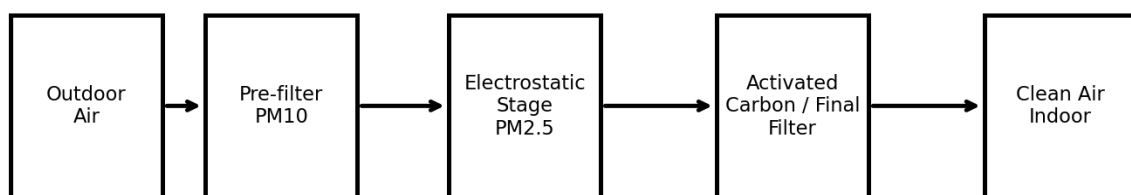
AIR CLEANING WINDOW PANEL

Complete Project & Discussion Summary

Consolidated from the project discussions

Concept • Design • Filtration • Sensors • Safety • Literature Review • Methodology • Construction • Evaluation

Air Cleaning Window Panel – Conceptual Airflow



Fan-driven airflow through integrated filtration stages

Conceptual illustration prepared for this summary; it is not a fabrication drawing.

Project theme: a compact window-integrated air-cleaning system intended to improve incoming indoor air quality while retaining the function of a normal window.

1. Executive Summary

The Air Cleaning Window Panel was developed as a compact, window-integrated air purification concept for buildings. The central idea is to place filtration and air-quality monitoring within a panel that can occupy a conventional window opening, allowing incoming air to be treated before entering the indoor environment.

The project evolved around a multi-stage filtration arrangement. A preliminary fabric filter was considered for larger suspended particles such as PM10, an electrostatic stage for finer particulate matter such as PM2.5, and a final activated-carbon/carbon-fiber-containing layer for odors and selected gaseous pollutants. A fan provides forced airflow through the system. Sensors and a microcontroller were included to monitor air quality, pressure drop and electrical behavior.

The discussions also addressed practical engineering issues: compactness, affordability, energy consumption, filter pressure drop, electrostatic safety, insulation and grounding, material selection, window-frame integration, and the need to evaluate the prototype experimentally.

2. Problem Statement

Conventional windows provide natural ventilation but do not actively remove airborne particulate matter and gaseous contaminants from incoming outdoor air. Standalone air purifiers can clean indoor air but occupy floor or table space and do not directly integrate with the ventilation opening. The project therefore investigates a hybrid approach in which a window panel performs both an architectural/window function and an air-cleaning function.

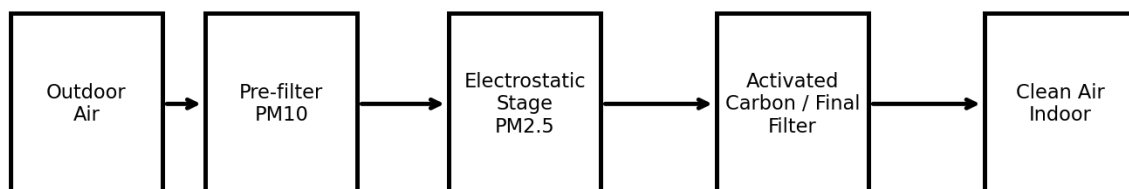
3. Project Objectives

The objectives discussed for the project can be consolidated as follows:

- Develop a compact air-cleaning device that can fit within a regular window frame.
- Reduce suspended particulate matter, particularly PM10 and PM2.5, from incoming air.
- Use activated carbon/carbon-based filtration to assist in controlling odors and gaseous pollutants.
- Integrate forced ventilation, sensing and filtration into one affordable and energy-conscious unit.
- Maintain safe operation of the electrostatic stage through insulation, grounding and controlled electrical design.
- Study airflow resistance, filter loading, air-quality response and overall prototype performance.

4. Overall System Concept

Air Cleaning Window Panel – Conceptual Airflow



Fan-driven airflow through integrated filtration stages

Figure 1. Conceptual airflow and filtration sequence.

5. Filtration System Development

5.1 Preliminary Fabric Filter

Cotton/fabric filtration was discussed as the first stage. Its role is to intercept larger particles and provide a preliminary barrier before air reaches the finer filtration stages. The discussion also considered the effect of dust accumulation and the formation of a filter cake, which can change filtration performance and pressure drop over time.

5.2 Electrostatic Filtration

The electrostatic stage was intended to target fine particulate matter, particularly PM_{2.5}. The concept involved a pair of electrically biased meshes/electrodes, with the detailed configuration evolving during the design discussion. A rubber bush/insulating separation concept was considered, while the separate insulating spacer was later removed from the proposed arrangement. Nichrome and related high-voltage components were discussed as part of the electrostatic separator concept.

5.3 Activated Carbon / Final Filter

Activated carbon was selected because of its adsorption capability for odors and various gaseous contaminants. Carbon fiber was also considered as part of the final filtration layer. The final fabric layer provides an additional physical barrier and helps protect the downstream side of the system.

6. Proposed Layer Arrangement

Conceptual Layer Arrangement / Exploded Sequence

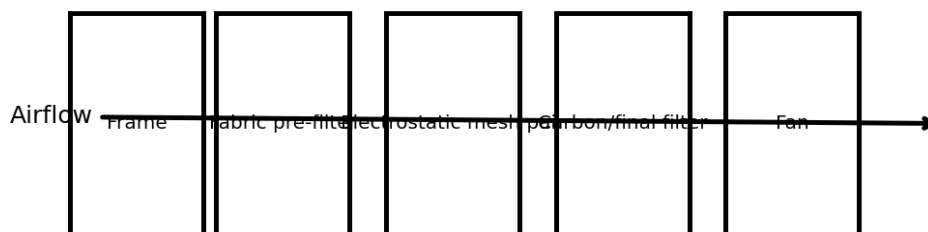


Figure 2. Conceptual layer sequence. Exact dimensions and final order should follow the validated prototype drawing.

The project discussions produced the following core three-stage concept:

Layer 1: preliminary fabric filter for larger particles/PM₁₀.

Layer 2: electrostatic mesh/electrode stage intended for PM_{2.5} capture.

Layer 3: final fabric/carbon-containing filter for residual particles, odor and selected gaseous pollutants.

7. Airflow and Ventilation

An external fan was included to drive air through the filtration stages. Forced airflow was selected to make the system less dependent on naturally available pressure differences across the window. The design therefore has to

balance clean-air delivery against fan energy use and pressure losses through the filters.

A key performance issue identified during the discussions was pressure drop. As filters load with dust, resistance to airflow can increase. This makes pressure monitoring useful and creates a basis for determining when cleaning or filter replacement may be required.

8. Sensors, Electronics and Control

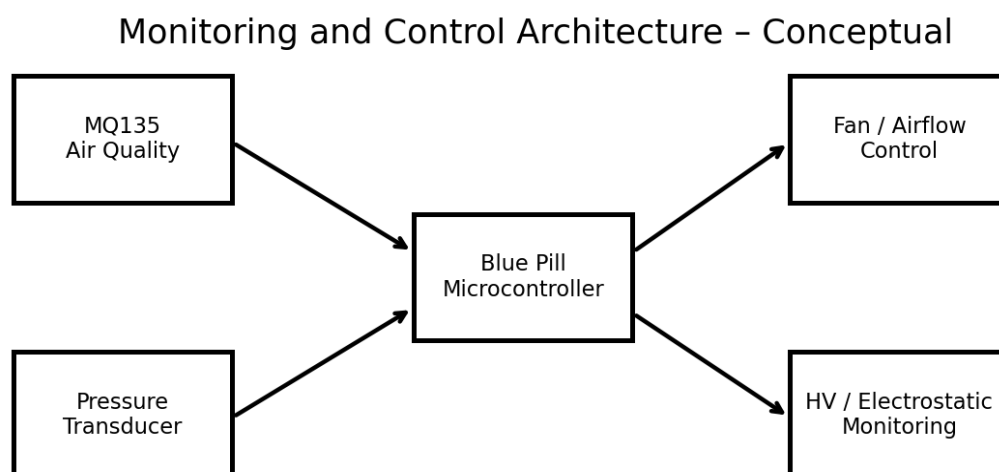


Figure 3. Conceptual monitoring and control architecture.

MQ135 air-quality sensor: discussed for monitoring changes in air quality and gaseous pollutants. It is useful as a relative/indicative air-quality sensor but should not be treated as a laboratory-grade pollutant analyzer without calibration.

Blue Pill microcontroller: proposed as the central controller for sensor acquisition, monitoring and possible control functions.

Pressure transducer/transmitter: intended to monitor pressure conditions across the filtration system and help identify increasing resistance or filter loading.

Current transducer: discussed for monitoring electrical current associated with the electrostatic system, supporting system monitoring and safety checks.

Power supply and wiring: required to provide regulated power to the control electronics, fan and electrostatic subsystem.

9. Electrostatic Safety Considerations

Safety was one of the major design concerns because the electrostatic stage requires elevated voltage. The discussions specifically considered the possibility of electric shock, sparking and accidental contact. The proposed safety approach included proper insulation, electrical separation, grounding, enclosed conductive parts, controlled high-voltage generation and safe routing of wiring.

A step-up transformer/high-voltage supply was discussed as part of the electrostatic system. The final prototype should only use a properly rated, current-limited and enclosed high-voltage supply with appropriate clearances and insulation. The panel should prevent user access to energized parts during normal operation.

10. Main Materials and Components Discussed

Category	Discussed item	Purpose
Frame	Wooden frame / aluminium structural element	Window integration and structural support
Filtration	Cotton/fabric filter	Preliminary particle filtration
Filtration	Electrostatic mesh/electrode stage	Fine particulate capture concept
Filtration	Activated carbon / carbon fiber	Adsorption of odors and selected gases
Air movement	External fan	Forced airflow through the panel
Sensing	MQ135	Indicative air-quality/gas sensing
Control	Blue Pill microcontroller	Data acquisition and control
Monitoring	Pressure transducer/transmitter	Pressure drop/filter loading monitoring
Electrical	Current transducer	Electrostatic electrical monitoring
HV system	Step-up transformer / HV supply	Electrostatic field generation
Electrodes	Nichrome and conductive mesh concepts	Electrostatic separator components
Safety	Insulation, grounding, rubber bush concept	Electrical isolation and safety

11. Frame and Mechanical Integration

The project was repeatedly framed around a compact panel that can occupy a regular window opening. A wooden frame and aluminium structural elements were discussed. The mechanical design has to accommodate the filters, electrostatic stage, fan, sensors, wiring and access for cleaning or replacement while maintaining adequate structural rigidity.

An exploded engineering drawing with main frame dimensions was also identified as a useful project deliverable. The exact dimensions were not finalized in the discussion, so the conceptual figures in this document are intentionally not presented as fabrication drawings.

12. Research and Literature Review Themes

The literature-review discussions covered several technical areas relevant to the project:

- **PM2.5 and indoor particulate matter:** health and indoor-air-quality significance of fine particles and the need for effective removal mechanisms.
- **VOC adsorption using activated carbon:** adsorption mechanisms and the usefulness of porous carbon materials for gaseous contaminants and odors.
- **Fabric filtration:** particle capture, filter-cake formation and the relationship between filtration efficiency and pressure drop.
- **HEPA filtration:** high-efficiency particulate removal as a benchmark and the associated pressure-drop considerations.
- **MXene nanosheets:** advanced material approaches for particulate filtration and functional filtration media.
- **Hybrid electrostatic filtration:** including the HYBRYDA+ concept discussed as an example of combining electrostatic and mechanical filtration mechanisms.
- **Natural ventilation:** window ventilation, airflow behavior and the limitations of relying on untreated outdoor air.
- **Trickle vents:** compact ventilation openings and their relevance to integrating ventilation with building envelopes.
- **Airflow and pressure drop:** the need to evaluate how filter resistance affects ventilation and fan requirements.

13. Research Methodology Discussed

The methodology can be organized into the following sequence:

1. Define the indoor-air-quality problem and functional requirements.
2. Review particulate and gaseous pollutant removal mechanisms.
3. Select the filtration stages and supporting materials.

4. Develop the window-panel mechanical layout and airflow path.
5. Integrate the fan, sensors, microcontroller, power supply and electrostatic subsystem.
6. Construct the prototype using the selected frame and filtration materials.
7. Test airflow, pressure behavior, electrical operation and air-quality response.
8. Compare before/after air-quality observations and assess the overall effectiveness of the prototype.
9. Identify design limitations and opportunities for further optimization.

14. Construction Phase – Consolidated Summary

The construction concept involved fabricating a rigid window-compatible frame, mounting the filtration layers in sequence, providing a controlled airflow path and installing the fan. The sensing and control electronics were planned around the Blue Pill microcontroller. The electrostatic stage required particular attention to insulation, grounding, electrical clearance and enclosure.

The prototype should also provide access for filter inspection and replacement. A practical panel must be maintainable because particulate accumulation will progressively affect pressure drop and filtration performance.

15. Testing and Performance Parameters

The project discussions imply several measurable parameters that should be considered during evaluation:

- PM10 and PM2.5 concentration before and after filtration.
- Indicative gaseous/air-quality sensor response using the MQ135.
- Airflow rate or air-delivery performance.
- Pressure drop across the filter assembly.
- Fan power/energy consumption.
- Electrostatic-stage current and stable electrical operation.
- Filter loading and performance change with use.
- Noise and practical usability, where measurements are available.

16. Expected Project Contributions

The project brings together building ventilation, particulate filtration, adsorption, electrostatic separation, sensing and compact mechanical integration in one window-mounted concept. Its main contribution is the exploration of an air-cleaning system that treats incoming ventilation air while occupying the space of a window panel rather than requiring a separate floor-standing purifier.

17. Strengths of the Proposed Concept

- Space-efficient and directly integrated with the building opening.
- Multi-stage approach addresses different pollutant classes.
- Potential for real-time monitoring through sensors and a microcontroller.
- Replaceable/maintainable filtration components are possible.
- Can be developed as a relatively compact and energy-conscious prototype.

18. Limitations and Engineering Challenges

- Electrostatic filtration introduces high-voltage safety requirements.
- Filter pressure drop can reduce airflow and increase fan power demand.
- MQ135 response is broad and requires calibration for quantitative pollutant measurement.
- Activated carbon performance depends on material properties, pollutant concentration, contact time and saturation.
- A compact panel has limited filter area compared with larger commercial systems.

- Window integration requires attention to sealing, structural support, maintenance and weather exposure.
- Prototype results need controlled testing before claims of quantitative purification efficiency are made.

19. Possible Future Development

Future iterations can focus on optimizing filter area and pressure drop, improving sensor calibration, refining the electrostatic stage, reducing power consumption, improving acoustic performance, developing replaceable filter cartridges and creating a more compact enclosure. A future version could also incorporate a more accurate particulate sensor for direct PM2.5 measurement.

20. Consolidated Project Architecture

The overall concept discussed across the project can therefore be represented as:

Window opening → pre-filtration → electrostatic fine-particle treatment → carbon/final filtration → fan-assisted delivery → indoor space

with the **MQ135, pressure transducer and current monitoring** connected to the **Blue Pill microcontroller** for system observation and control.

21. Important Design Decisions Recorded During Discussion

- The project was defined as a window-integrated air-cleaning panel rather than a conventional standalone purifier.
- A three-layer filtration concept became the main arrangement.
- A preliminary cotton/fabric filter was retained for larger particles.
- An electrostatic mesh/electrode stage was retained for PM2.5-oriented filtration.
- Activated carbon/carbon fiber was included in the final filtration concept for odor/gaseous pollutant control.
- A separate insulating spacer was considered and later removed from the proposed design; electrical isolation remained a safety requirement.
- A rubber-bush concept was discussed for electrical/mechanical separation.
- An external fan was selected for forced airflow.
- MQ135, Blue Pill, current sensing and pressure sensing were incorporated into the monitoring concept.
- An engineering drawing with main frame dimensions was identified as a required design output.

22. Final Consolidated Understanding

The Air Cleaning Window Panel is a compact civil/building-services engineering prototype that combines a window frame with a multi-stage air-cleaning system. The project sits at the intersection of indoor environmental quality, ventilation, filtration, materials, electronics and building integration. The core research question is how effectively a compact window-mounted system can improve the quality of incoming air while maintaining acceptable airflow, energy use, safety and maintainability.

This summary consolidates the topics discussed in the available project history. It distinguishes established project concepts from areas where the exact prototype dimensions, final component specifications and quantitative test results had not yet been fixed.

Note on figures

The figures in this PDF are original conceptual diagrams created specifically for this summary. They are intended to communicate the system architecture and should not be used as fabrication drawings without detailed engineering validation.